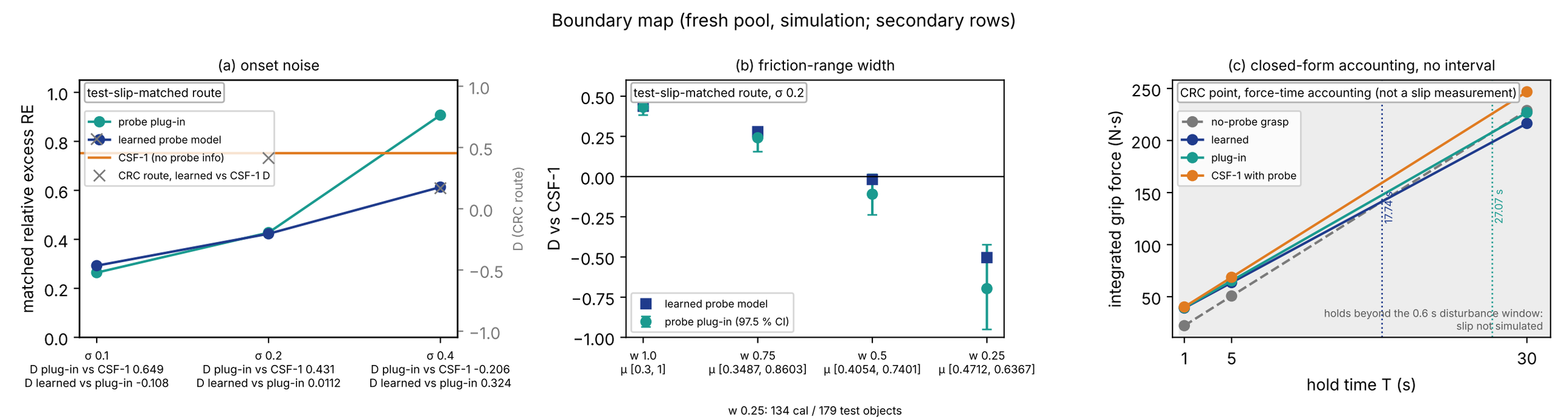
Boundary map (fresh pool, simulation; secondary rows)
(a) onset noise
test-slip-matched route
probe plug-in
learned probe model
CSF-1 (no probe info)
CRC route, learned vs CSF-1 D
matched relative excess RE
D (CRC route)
σ 0.1
D plug-in vs CSF-1 0.649
D learned vs plug-in -0.108
σ 0.2
D plug-in vs CSF-1 0.431
D learned vs plug-in 0.0112
σ 0.4
D plug-in vs CSF-1 -0.206
D learned vs plug-in 0.324
(b) friction-range width
test-slip-matched route, σ 0.2
D vs CSF-1
learned probe model
probe plug-in (97.5 % CI)
w 1.0
μ [0.3, 1]
w 0.75
μ [0.3487, 0.8603]
w 0.5
μ [0.4054, 0.7401]
w 0.25
μ [0.4712, 0.6367]
w 0.25: 134 cal / 179 test objects
(c) closed-form accounting, no interval
CRC point, force-time accounting (not a slip measurement)
no-probe grasp
learned
plug-in
CSF-1 with probe
integrated grip force (N·s)
17.47 s
27.07 s
holds beyond the 0.6 s disturbance window:
slip not simulated
hold time T (s)